FORGE — FDE ACADEMY · CONSOLIDATION · ADR-2 SIGN-OFF

One System, Signed Off

Integration · Cost · ADR-2 · W3 Handover

Wire Foundry + LangGraph + CI/CD + eval + telemetry into one live system · review W2 cost budget vs actual · lock the retrieval architecture in ADR-2 · clear the 30-min load test before Week-3 data-discovery

capstone integration

budget vs actual

ADR-2 sign-off

30-min load test

W3 readiness

SESSION MAP
What we cover today

Capstone integration sprint — 1
Foundry + LangGraph + CI/CD + eval + telemetry wired into one live system

Cost review across W2 — 2
Budget vs actual · rightsizing · the PTU vs PAYG decision, revisited

ADR-2 final sign-off — 3
Lock the retrieval architecture before Week-3 data-discovery begins

W2 → W3 handover readiness — 4
Every pod clears a 30-min load test and passes the eval gates

Sign-off & reference — 5
Capstone acceptance · glossary · what carries into Week 3

OUTCOMES
By the end, you will have…

One integrated system

The whole W2 stack running end-to-end: request in, grounded answer out, telemetry flowing.

A cost verdict

Budget vs actual for every W2 service, with rightsizing and a PTU/PAYG call.

ADR-2 signed

The retrieval architecture locked, with options, decision and consequences on record.

A load-tested build

Every pod clears a 30-min load test and the eval gate — proven, not hoped.

W3 readiness

A clean handover: what's done, what's locked, what W3 data-discovery starts from.

Capstone accepted

W2 capstone fully integrated & load-tested; ADR-2 finalised & signed.

TECHADEMY

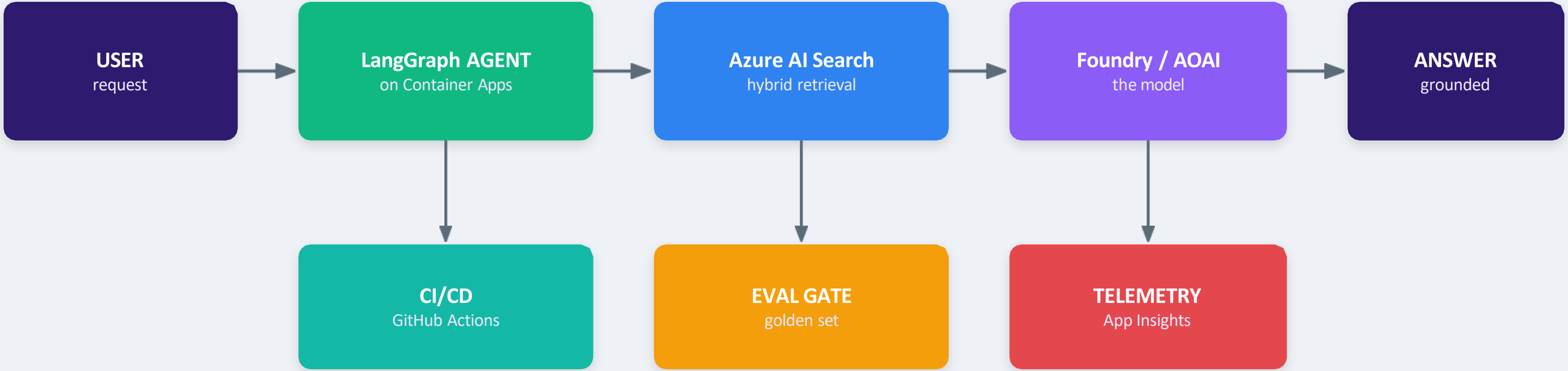
01

CAPSTONE INTEGRATION SPRINT

Wire it into one live system

THE WHOLE PICTURE

The Week-2 system, wired together



Cross-cutting: Managed Identity (Entra ID) · Cosmos DB state · blue/green revisions · OpenTelemetry

Integration means these stop being seven demos and become one system: a request flows through retrieval and the model to a grounded answer, while CI/CD ships it, the eval gate guards it, and telemetry watches it.

INTEGRATION CHECKLIST
What wires to what

Code → deploy

A push to main builds the image, runs the eval gate, and blue/green-deploys the agent to Container Apps.

Agent → retrieval

LangGraph calls Azure AI Search as a typed tool; grounded chunks come back before the model answers.

Deploy → gate

The eval gate on the golden set sits in the pipeline — a regression blocks the release.

Everything → telemetry

Every model and tool call emits gen_ai.* spans to Application Insights: tokens, p99, cost, per agent.

State → Cosmos

Conversation and checkpoints persist in Cosmos DB so the agent can resume after a crash.

Identity everywhere

Managed Identity (Entra ID) — the agent pulls images, reads secrets and calls services without passwords.

PROVE IT END-TO-END
The integration smoke test

One request, all the way through

1. Hit the deployed FQDN with a known question.
2. Confirm LangGraph routed and called the retrieval tool.
3. Confirm Azure AI Search returned grounded chunks.
4. Confirm the answer cites those chunks (faithful).
5. Confirm the trace in App Insights shows tokens + p99.
6. Confirm state was written to Cosmos DB.

smoke.sh

```
# smoke test the live system
curl -s https://bank-agent.<env> \
  .azurecontainerapps.io/ask \
  -H 'content-type: application/json' \
  -d '{"q":"What is the KYC hold policy?"}'

# expect: grounded answer + citations
{
  "answer": "...per policy KYC-14...",
  "citations": ["kyc-policy#s3"],
  "trace_id": "9f2c...",
  "tokens": {"in": 812, "out": 143},
  "latency_ms": 1180
}
# then: open trace_id in App Insights
```

02

COST REVIEW ACROSS W2

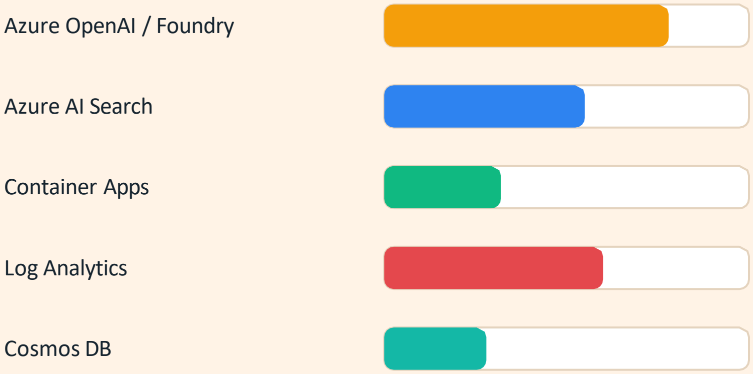
Budget vs actual · rightsizing

WHERE DID THE MONEY GO?
Budget vs actual

Read it in Cost Management

- Tag every W2 resource by feature so cost stops at the feature, not the resource group.
- Set an Azure Cost Management budget with alerts at 50 / 80 / 100% of the monthly cap.
- Compare actual vs budget by service — the variance, not the total, tells the story.
- Remember the ~15–40% non-token overhead: support, networking, Log Analytics, egress.

Typical W2 variance



Log Analytics is the classic surprise — verbose default logging bills as monitoring, not as 'AI cost'.

TRIM THE WASTE
Rightsizing the W2 services

Container Apps replicas

Match min/max replicas to real traffic. Scale-to-zero the dev/test app; keep 1 warm replica in prod.

AI Search tier

Right-size the tier to index size + QPS. Don't run S2 for a demo-scale index; scale up only when recall/latency needs it.

Model tiering

Route classification / routing to a nano tier; reserve the frontier model for the hard 20%.

Kill zombies

Delete idle fine-tunes, forgotten test deployments and unused PTU reservations — silent, recurring spend.

Rightsizing is measured, not guessed: change one lever, watch the ProvisionedUtilization / replica / recall metric, then keep or revert.

THE DECISION, REVISITED
PTU vs PAYG — now with real data

What changed since Week 1

- In Week 1 the PTU/PAYG call was a guess. Now you have 2 weeks of real telemetry.
- Pull P95 hourly token throughput from Application Insights — the true input to the decision.
- PTU wins above ~50% sustained utilisation (~150–200M GPT-4o tokens/month).
- Below that line, PAYG's flexibility wins — don't reserve capacity you won't fill.
- Hybrid is usually the answer: PTU for the steady baseline, PAYG for burst + dev.

The W2 verdict (worked example)

Measured util (P95)	**38%**
Break-even util	**~50%**
Monthly tokens	**~90M**
Verdict	**Stay PAYG for now**

Re-check the call when…

- Sustained utilisation crosses ~50% for a full month.
- A go-live pushes steady volume past ~150M tokens/month.
- Tail latency (p99) needs the guaranteed throughput PTUs give.

SERVICE BY SERVICE

The W2 cost review, on one page

Service	Budget	Actual	Action
Azure OpenAI / Foundry	$1,200	**$1,540**	Enable prompt caching; cap max_tokens
Azure AI Search	$700	**$720**	Right tier; revisit at W3 index growth
Container Apps	$300	**$180**	Scale-to-zero dev; keep 1 warm in prod
Log Analytics	$150	**$260**	Sample telemetry; drop verbose logs
Cosmos DB	$200	**$150**	Autoscale RU/s to traffic
Total	$2,550	**$2,850**	~12% over — caching + sampling close it

Editorial layer: these figures are an illustrative template — replace with your Cost Management export. The pattern is what matters: name the variance, name the action.

TECHADEMY

03

ADR-2 FINAL SIGN-OFF

Lock retrieval before W3

THE INSTRUMENT
What an ADR is — and why lock it now

Context
the forces at play — value-neutral facts

Decision
"We will..." — active voice, one choice

Consequences
what results — the good AND the bad

Status
proposed / accepted / superseded

ADR-2 · CONTEXT
Why we're deciding retrieval now

Context

- The W2 agent must ground its answers in the bank's document corpus.
- Vector-only RAG in the lab missed exact terms (policy numbers, codes) and lost context across chunks.
- Week-3 data-discovery (semantic layers, knowledge graphs) needs a settled retrieval baseline to build on.
- The choice must be enterprise-fit on Azure, not a notebook demo.

Decision drivers

Retrieval accuracy

recall@k and faithfulness on the golden set

Cost per query

tokens + any premium ranking / reasoning

Latency (p95)

must survive the 30-min load test

Explainability

citations an auditor can follow

Enterprise fit

managed, secure, on Azure

ADR-2 · CONSIDERED OPTIONS
Four ways to retrieve — scored honestly

Option	Accuracy	Cost	Notes
Vector-only RAG	Medium	Low	Cheapest; misses exact terms, loses context across chunks
Hybrid + RRF + semantic	High	Medium	BM25 + vector fused, then re-ranked — MS benchmarked default
Vectorless / PageIndex	Highest	High	~98–99% but an LLM reasons per node — cost uncontrolled
Knowledge graph	High*	High	Great for linked entities; heavy to build — a W3 topic

MADR discipline: name the losers and why. The winner isn't the most accurate at any price — it's the best accuracy we can afford, on Azure, within p95.

ADR-2 · DECISION
Hybrid retrieval on Azure AI Search

L1 Retrieve

BM25 keyword // vector (HNSW)
run in parallel

→

L2 Fuse

RRF fusion (k=60)
merge the two lists

→

L3 Re-rank

Semantic ranker (top 50)
cross-encoder re-score

Decision

- Hybrid (keyword + vector) ⟶ RRF ⟶ semantic ranker on Azure AI Search.
- Chunking: 512 tokens with ~25% overlap (Text Split skill).
- This is Microsoft's benchmarked default (vector_semantic_hybrid).

Why it beats the alternatives

- Vector finds meaning; keyword catches exact codes & policy numbers.
- RRF fuses without needing same-scale scores.
- Semantic re-rank lifts the truly relevant chunk to the top.

Y-STATEMENT

In the context of **grounding the W2 agent before W3 data-discovery**, facing **an accuracy-vs-cost trade-off**, we chose **hybrid + RRF + semantic ranking on Azure AI Search**, to achieve **benchmark-leading relevance out of the box**, accepting **the added cost of the semantic ranker + a dual index.**

ADR-2 · CONSEQUENCES
What we accept — and the sign-off

Positive

- Best relevance we can buy on Azure, with citations for audit.
- One managed service — no bespoke retrieval stack to operate.
- A settled baseline W3 can build semantic layers on.

Negative (accepted)

- Semantic ranker is a premium, billed feature — watch its cost.
- Two indexes (keyword + vector) to build and keep in sync.
- Not a knowledge graph — linked-entity questions wait for W3.

ADR-2 · STATUS: ACCEPTED

DECISION	DECIDERS	DATE	SUPERSEDES
Hybrid + RRF + semantic on AI Search	Architect + trainer + reviewer	Week-2, signed	— (extends ADR-1)

TECHADEMY

04

W2 → W3 HANDOVER READINESS

Clear the gates before W3

PROVE IT HOLDS

The 30-minute load test

It's a gate, not a report

- Azure Load Testing runs managed JMeter/Locust against the live endpoint.
- Define fail criteria — the test passes or fails, and that result is the gate.
- Watch tail latency (p95/p99), not the average — that's what real users feel.
- Read client metrics (RPS, latency, errors) and server metrics (CPU, scaling) together.
- Auto-stop on a high error rate protects you from a runaway, costly run.

loadtest.yaml

```yaml
# loadtest.yaml — fail criteria = gate
testName: agent-30min
engineInstances: 2
duration: 1800          # 30 minutes
loadPattern:
  virtualUsers: 500
failureCriteria:
  - avg(response_time_ms) > 2000
  - percentage(error) > 1
-   p99(response_time_ms) > 4000
autoStop:
  errorRate: 90         # safety cut-out
# run in CI:
#  az load test create ...
#  -> pass/fail sets the pipeline status
```

EVERY POD MUST CLEAR
The handover gate matrix

Load test

30 min at target load; p95 < 2s, errors < 1%. A fail blocks handover.

Eval gate

Golden-set score no more than 2% below baseline — faithfulness holds under change.

Telemetry live

Tokens, p99 and a quality signal visible in App Insights, wired to alerts.

Rollback rehearsed

One-command traffic flip to last-good, proven in a game-day.

Release documented

Prompt version pinned; model card auto-generated for the release.

ADR-2 signed

Retrieval architecture locked and referenced by the pod's design.

All six green = the pod may hand over to Week 3. Any red = it stays in W2 until fixed. No exceptions, no 'just this once'.

THE HANDOVER
W2 → W3 readiness checklist

- ✓ The whole system passes the end-to-end smoke test from the deployed endpoint.
- ✓ Every pod cleared the 30-minute load test with p95/error criteria met.
- ✓ The eval gate is live and has visibly blocked at least one bad change.
- ✓ Telemetry (tokens · p99 · quality) flows to App Insights with alerts set.
- ✓ The rollback runbook is pinned and rehearsed; last-good revision is known.
- ✓ Cost review done: variances named, owners and actions assigned.
- ✓ ADR-2 accepted and signed; retrieval architecture is locked.
- ✓ W3 scope is clear: data-discovery starts from the ADR-2 baseline.

WHAT W3 STARTS FROM
Next: data-discovery

Semantic layers

- A shared meaning layer over raw data: canonical entities, metrics and relationships.
- Lets the agent reason over 'what things mean', not just which chunk matched.
- Builds on ADR-2's retrieval — it enriches, it doesn't replace it.

Knowledge graphs

- Entities as nodes, relationships as edges — perfect for linked-entity questions.
- Answers 'which accounts connect to this flagged customer?' that vectors can't.
- The heavy option ADR-2 deliberately deferred — now the W3 exploration.

The thread holds: retrieval is locked (ADR-2), the system is load-tested, and W3 layers meaning and relationships on top of a stable base.

TECHADEMY

05

SIGN-OFF & REFERENCE

Close out Week 2

WEEK-2 CAPSTONE
Signed off — the two outcomes

Fully integrated & load-tested

- One live system: Foundry + LangGraph + CI/CD + eval + telemetry.
- Smoke test green end-to-end; grounded answers with citations.
- 30-min load test passed: p95 < 2s, errors < 1%.

ADR-2 finalised & signed

- Retrieval locked: hybrid + RRF + semantic ranker on AI Search.
- Options weighed, consequences recorded, deciders named.
- Status: Accepted — the W3 baseline is set.

Week 2 complete

The system is integrated and proven under load; the retrieval decision is on record. Week 3 begins from a stable, signed baseline.

GLOSSARY · 1 OF 3
Integration & Azure

Azure AI Foundry Azure's platform for building & hosting AI models/agents.

LangGraph Graph-based framework for stateful multi-step agents.

Container Apps Serverless container hosting; runs the deployed agent.

Managed Identity Passwordless Azure identity (Entra ID) for service auth.

Cosmos DB Managed NoSQL store; holds agent state & checkpoints.

OpenTelemetry Vendor-neutral standard for traces, metrics, logs.

gen_ai.* spans Standard telemetry attributes for LLM calls.

Blue/green Two revisions; shift traffic old→new for safe releases.

Smoke test One end-to-end request proving the system is wired.

Application Insights Azure Monitor's APM; where telemetry lands.

ACR Azure Container Registry — private image store.

Integration sprint Wiring separate services into one working system.

GLOSSARY · 2 OF 3
Retrieval & ADR

ADR Architecture Decision Record — one decision, on the record.

MADR Markdown ADR template with drivers & considered options.

Y-statement One-line ADR summary: context / decision / trade-off.

Superseded Status of an ADR replaced by a newer one (not deleted).

Hybrid search Keyword (BM25) + vector search combined.

BM25 Classic keyword relevance ranking; catches exact terms.

Vector search Embedding similarity; captures meaning / paraphrase.

HNSW Graph index for fast approximate vector search.

RRF Reciprocal Rank Fusion — merges ranked lists (k=60).

Semantic ranker Cross-encoder that re-scores top results by meaning.

Chunking Splitting docs into passages (e.g. 512 tok, 25% overlap).

recall@k Share of relevant items found in the top-k results.

Faithfulness Whether an answer is grounded in retrieved sources.

Knowledge graph Entities + relationships; a W3 exploration topic.

GLOSSARY · 3 OF 3
Cost, load-test & handover

Budget vs actual Planned spend compared to what was billed.

Azure Cost Management Azure's tool for budgets, alerts & cost analysis.

Rightsizing Matching allocated capacity to real demand.

PTU Provisioned Throughput Unit — reserved model capacity.

PAYG Pay-as-you-go — per-token billing, no commitment.

ProvisionedUtilization Metric showing how much PTU capacity is used.

Break-even Utilisation where PTU beats PAYG (~50%).

Azure Load Testing Managed JMeter/Locust load-testing service.

Fail criteria Thresholds that make a load test pass or fail.

p95 / p99 95th / 99th-percentile latency — the slow tail.

RPS Requests per second — throughput under load.

Virtual users Simulated concurrent users in a load test.

Auto-stop Safety cut-out that halts a failing/costly run.

Gate matrix The set of checks a pod clears to hand over to W3.

REFERENCE
Sources & further reading

Architecture Decision Records (Nygard, templates)	adr.github.io
MADR — Markdown ADR template	adr.github.io/madr
Hybrid search in Azure AI Search	learn.microsoft.com
Semantic ranking in Azure AI Search	learn.microsoft.com
Azure Load Testing overview	learn.microsoft.com
Define load-test fail criteria	learn.microsoft.com
Create & manage budgets (Cost Management)	learn.microsoft.com
Provisioned throughput (PTU) concepts	learn.microsoft.com

TECHADEMY

RECAP

Integrated, costed, decided, ready

- Integration: seven services became one live system — smoke-tested end-to-end.
- Cost: budget vs actual reviewed, rightsized, and PTU vs PAYG re-decided on real data.
- ADR-2: hybrid + RRF + semantic retrieval on Azure AI Search — accepted and signed.
- Handover: every pod clears a 30-min load test and the eval gate before Week 3.
- Week 3 begins from a locked, load-tested baseline: semantic layers & knowledge graphs.